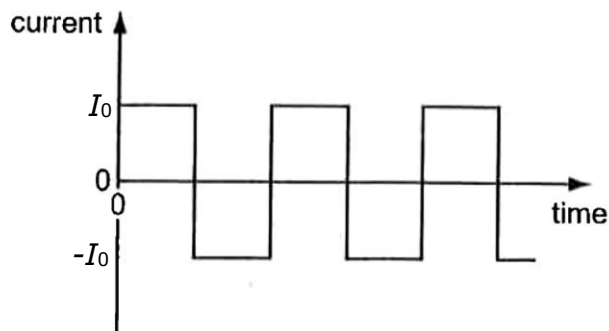


26 The graph shows an alternating current with a square waveform.



The peak value of the current is  $I_0$  and the average power dissipated in the circuit is given by  $P$ .

What is the new average power dissipated if the peak value is now doubled to  $2I_0$ ?

- A**  $P$                       **B**  $1.4P$                       **C**  $2P$                       **D**  $4P$

Ans: D

The  $I_{rms}$  of the square waveform is  $I_0$ .

$$\text{(working: } I_{rms} = \sqrt{\frac{I_0^2 \times \frac{T}{2} + I_0^2 \times \frac{T}{2}}{T}} = I_0 \text{)}$$

The average power is given by

$$P = I^2 R = (I_0)^2 R$$

**27** In a photoelectric effect experiment, a metallic surface X in an evacuated tube is illuminated with